

Minimodule 5. Sets and combinatorics

List of exercises (from 7th edition of the book)

Section 2.2 Exercises 3, 25, 20, 27

Section 6.1 3, 7

Section 6.3 8, 11, 21, 23, 38

Section 6.4 3

Section 6.1 8, 27

Section 6.3 19

3. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

- a) $A \cup B$. b) $A \cap B$.
c) $A - B$. d) $B - A$.

25. Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find

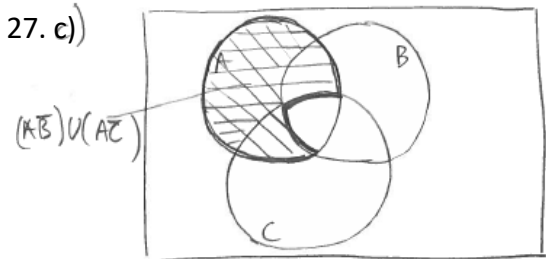
- a) $A \cap B \cap C$. b) $A \cup B \cup C$.
c) $(A \cup B) \cap C$. d) $(A \cap B) \cup C$.

In exercise 20 you can use Venn diagrams

20. Show that if A and B are sets with $A \subseteq B$, then

- a) $A \cup B = B$.
b) $A \cap B = A$.

Example for 27. c))



27. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .

- a) $A \cap (B - C)$ b) $(A \cap B) \cup (A \cap C)$
c) $(A \cap \overline{B}) \cup (A \cap \overline{C})$

3. A multiple-choice test contains 10 questions. There are four possible answers for each question.

- a) In how many ways can a student answer the questions on the test if the student answers every question? 4^{10}
b) In how many ways can a student answer the questions on the test if the student can leave answers blank? 5^{10}

7. How many different three-letter initials can people have? English alphabet 26^3

3. How many permutations of $\{a, b, c, d, e, f, g\}$ end with a ? $6! = P(6,6)$

8. In how many different orders can five runners finish a race if no ties are allowed? $5! = P(5,5)$

11. How many bit strings of length 10 contain

- a) exactly four 1s? $C(10,4)$
b) at most four 1s? $C(10,0) + C(10,1) + C(10,2) + C(10,3) + C(10,4)$
c) at least four 1s? Count all combinations with 4 or more, or adjust for all with less, 848
d) an equal number of 0s and 1s? $C(10,5)$

21. How many permutations of the letters *ABCDEFGH* contain

- a) the string *BCD*? $P(5,5)=5!$
- b) the string *CFGA*? $4!$
- c) the strings *BA* and *GF*? $5!$
- d) the strings *ABC* and *DE*? $4!$
- e) the strings *ABC* and *CDE*? $3!$
- f) the strings *CBA* and *BED*? 0

23. How many ways are there for eight men and five women to stand in a line so that no two women stand next to each other? [*Hint*: First position the men and then consider possible positions for the women.] $P(9,5)$

38. How many ways are there to select 12 countries in the United Nations to serve on a council if 3 are selected from a block of 45, 4 are selected from a block of 57, and the others are selected from the remaining 69 countries? $C(45,3) + C(57,4) + C(69,5)$

3. Find the expansion of $(x + y)^6$. $((x + y)^6 = x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6)$

8. How many different three-letter initials with none of the letters repeated can people have? English alphabet, $P(26,3)$

8a. How many different three-letter initials with one of the letters repeated can people have? English alphabet, $C(3,2)P(26,2)$

27. A committee is formed consisting of one representative from each of the 50 states in the United States, where the representative from a state is either the governor or one of the two senators from that state. How many ways are there to form this committee? 3^{50}

19. A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes

- a) are there in total? 2^{10}
- b) contain exactly two heads? $C(10,2)$
- c) contain at most three tails? $C(10,0) + C(10,1) + C(10,2) + C(10,3)$
- d) contain the same number of heads and tails? $C(10,5)$